

Minesweeper Testing Documentation

Software Engineering Project
Manual functional and edge case testing
September 18, 2026

1. Test Scope

Testing focused on the main game states and boundary conditions for the 10 x 10 Minesweeper board. The checks covered the selectable mine range, flag limits, first-click safety, win and loss behavior, timer behavior, and basic input handling.

The first-click safety rule was tested from the center of the board, from an edge, and from a corner. In each case, the selected cell and the neighboring cells that existed around it were safe.

2. Completed Manual Tests

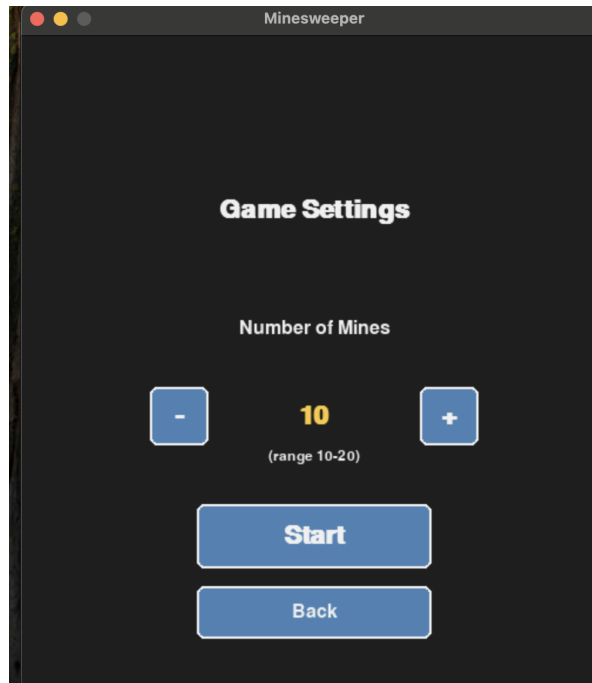
ID	Test	Procedure	Expected / observed result	Status	Evidence
TC-01	Minimum mine boundary	Set mine count to 10 and continue pressing the minus button.	The value stays at 10.	Pass	Screenshot 1
TC-02	Maximum mine boundary	Set mine count to 20 and continue pressing the plus button.	The value stays at 20.	Pass	Screenshot 2
TC-03	Maximum flag count	Start a 20-mine game, place 20 flags, then attempt another flag.	The counter reaches Mines: 0 and no extra flag is placed.	Pass	Screenshot 3
TC-04	Flags before first reveal	Place all available flags before making the first left-click.	Flags can be placed while the board remains unrevealed.	Pass	Screenshot 4
TC-05	Mines: 0 is not a win	Use all available flags without revealing every safe cell.	The game continues. Mines: 0 alone does not trigger a win.	Pass	Screenshot 5
TC-06	First-click safety	Start new games and make the first reveal in the center, on an edge, and in a corner.	The first cell and its available neighbors are safe.	Pass	Manual test
TC-07	Win without flags	Reveal all safe cells without placing flags.	The game reports You Win even with unused flags remaining.	Pass	Screenshot 6
TC-08	Standard win	Flag mines and reveal all safe cells.	The game reports You Win and the board stops accepting play input.	Pass	Screenshot 7
TC-09	Loss on mine click	Reveal a mine during play.	Game Over appears and the mines are displayed.	Pass	Screenshot 8
TC-10	Large timer value	Allow the timer to continue for an extended period.	The timer continues to display a valid value without affecting the HUD.	Pass	Screenshot 5

3. Edge Case and Regression Checklist

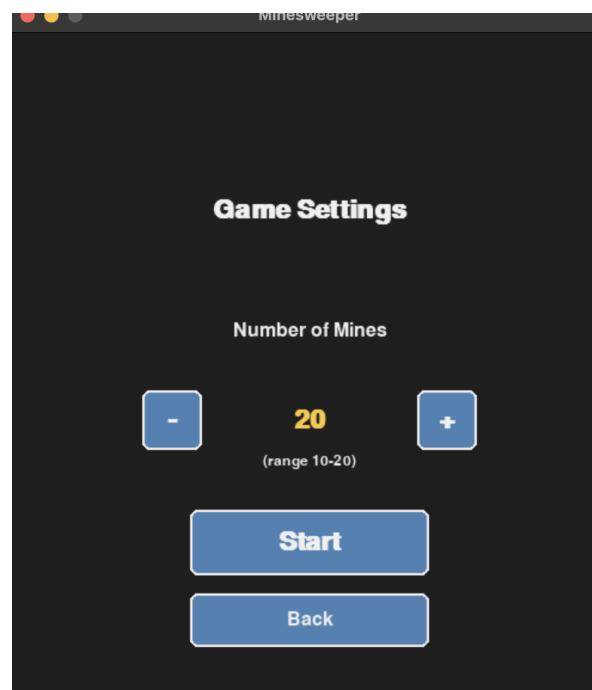
These cases were also considered during testing and code review. They can be repeated after future changes to the game.

ID	Edge case	Check	Expected behavior
EC-01	Unflag at flag limit	Reach the flag limit, then right-click an existing flag.	The existing flag can still be removed.
EC-02	Left-click a flagged cell	Flag a hidden cell, then left-click it.	The cell remains hidden and flagged.
EC-03	Flag a revealed cell	Reveal a safe cell, then right-click it.	No flag is added.
EC-04	Repeated reveal	Click the same revealed cell several times.	The board state does not change incorrectly.
EC-05	Chord with correct flags	On a revealed number, flag the correct neighboring mines and click the number.	Remaining safe neighboring cells reveal.
EC-06	Chord with too few flags	Click a revealed number before enough neighboring flags are placed.	No chord reveal occurs.
EC-07	Chord with incorrect flags	Place the correct number of flags but put one on a safe cell, then chord.	An unflagged mine can be exposed and the game ends normally.
EC-08	Post-game input	After a win or loss, try to reveal or flag cells.	The board does not change.
EC-09	Timer after game over	Record the time at win or loss and wait.	The displayed time remains fixed.
EC-10	Reset and menu controls	Use Reset, Menu, R, M, and Escape in their supported states.	The correct game or menu state is loaded.
EC-11	Flood-fill with a flag	Flag a safe cell in an area that would normally auto-reveal, then open the zero region.	Flood-fill skips the flagged cell.
EC-12	Non-board mouse input	Click the HUD, window edges, or use an unsupported mouse button.	No board cell changes.
EC-13	Out-of-range board index	Call board methods directly with an invalid row or column in a unit test.	Record current behavior and decide whether explicit bounds checking is needed.

4. Screenshot Evidence

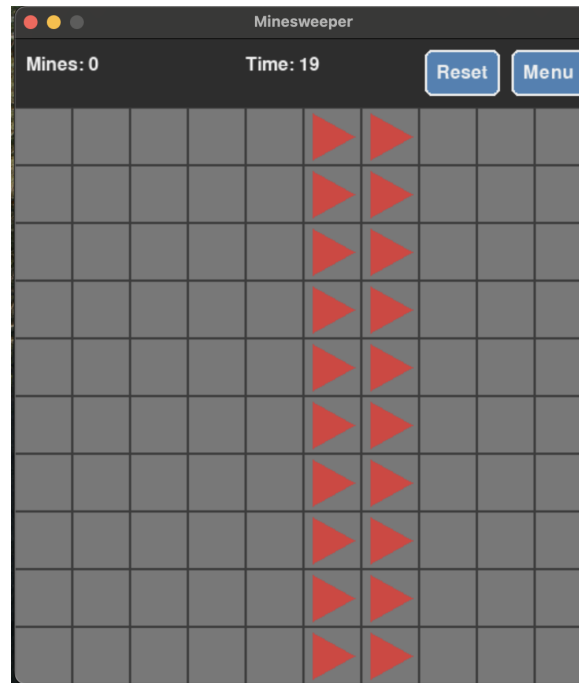


Screenshot 1. Minimum mine setting (10).

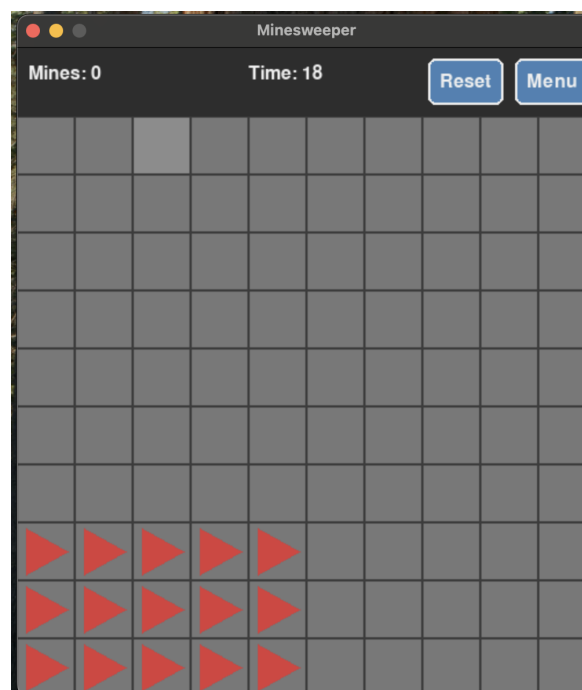


Screenshot 2. Maximum mine setting (20).

4. Screenshot Evidence (continued)



Screenshot 3. Maximum flag count on a 20-mine game.

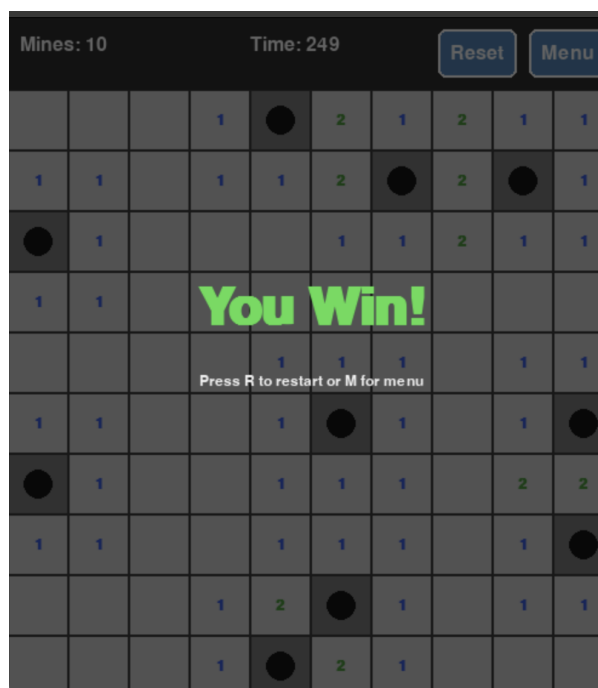


Screenshot 4. All available flags placed before the first reveal.

4. Screenshot Evidence (continued)

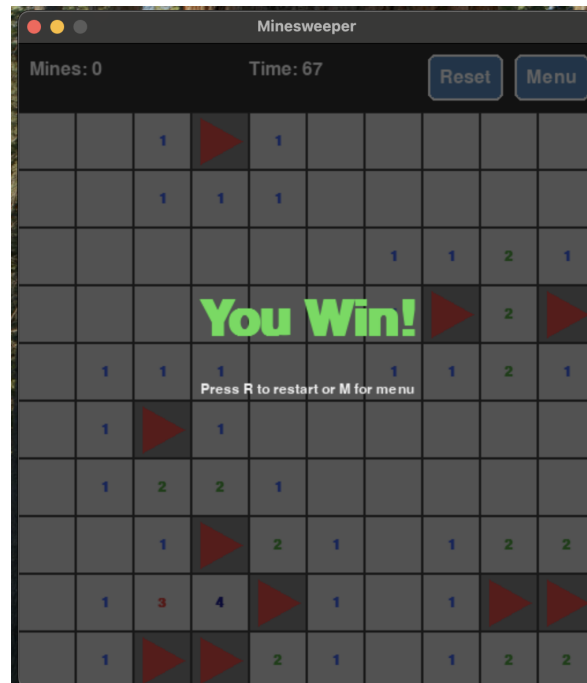


Screenshot 5. Mines: 0 without a win, with a large timer value.

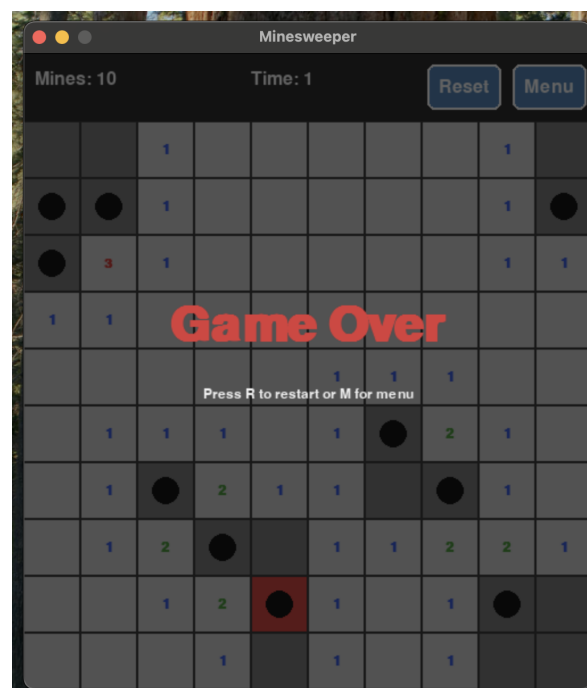


Screenshot 6. Win reached without placing flags.

4. Screenshot Evidence (continued)



Screenshot 7. Standard winning state.



Screenshot 8. Losing state after selecting a mine.

4. Screenshot Evidence (continued)



Screenshot 9. In-progress board used during manual gameplay testing.

5. Result

The completed manual tests listed in Section 2 passed during this test session. The screenshots in Section 4 record the main boundary, flag, win, loss, timer, and gameplay states used during testing.